

# SaiKiran Tedla

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Address: Toronto, Ontario

## Education

- York University, Toronto, Canada** Aug 2022–Aug 2026  
*PhD in Computer Science* GPA: A+  
PhD Thesis: Demosaicing for New Sensor Patterns  
Supervisor: [Dr. Michael Brown](#)
- York University Toronto, Canada** Aug 2020–Jul 2022  
*Master of Science in Computer Science* GPA: A  
MS Thesis: [Analyzing Color Imaging Failure on Consumer Grade Cameras](#)  
Supervisor: [Dr. Michael Brown](#)
- University of Colorado Springs, Colorado, USA** Aug 2015–May 2019  
*Bachelor of Science in Computer Engineering* GPA: 3.99/4.0  
**Outstanding Undergraduate Student**

## Experience

- Schmidt AI Postdoctoral Fellow, University of Toronto** Sept 2026–Present  
• **Computational Imaging Group**, [Dr. David Lindell](#) and [Dr. Kyros Kutulakos](#) *Toronto, Canada*  
• Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship; exploring generative re-photography and camera editing.
- Research Student, York University** May 2020–Aug 2026  
• **Computational Vision and Imaging Lab**, [Dr. Michael Brown](#) *Toronto, Canada*  
• Research on computational imaging: color correction, exposure, focus, compression, and demosaicing.
- Visiting Research Student, University of Toronto** Feb 2025–Aug 2026  
• **Computational Imaging Group**, [Dr. David Lindell](#) and [Dr. Kyros Kutulakos](#) *Toronto, Canada*  
• Utilizing video diffusion models for image editing and working on low-light 3D reconstruction problems
- Research Intern, Sony AI** July 2025–Mar 2026  
• **Imaging and Sensing**, [Dr. Feiran Li](#) and [Dr. Daisuke Iso](#) *Tokyo, Japan*  
• Learning HDR reconstruction methods
- Research Intern, Samsung AI Center Toronto** Mar 2025–July 2025  
• **Computational Photography**, [Dr. Michael Brown](#) and [Dr. Mahmoud Affi](#) *Toronto, Canada*  
• Improving color correction within the camera pipeline with a focus on colorimetric accuracy
- Research Intern, Adobe** May 2024–Jan 2025  
• **NextCam**, [Dr. Shumian Xin](#), [Dr. Cecilia Zhang](#), and [Dr. Zhoutong Zhang](#) *San Jose, USA*  
• Implementing video diffusion models and 3D representations for the task of refocusing images
- Research Intern, Samsung AI Center Toronto** Aug 2022–Aug 2023  
• **Computational Photography**, [Dr. Michael Brown](#) *Toronto, Canada*  
• Learning demosaicing for various sensor patterns and raw reconstruction (Graphics2RAW)
- Computer Vision Consultant, Scanwell Health** Apr 2021–Jan 2022  
• **Color Measurement** *Los Angeles, USA*  
• Developed custom color correction algorithms for medical imaging applications

## Research Student / System Administrator, UCCS

June 2015–Jan 2021

- Vision and Security Technology (VAST) Lab, [Dr. Terrance Boulton](#) *Colorado Springs, USA*
- Built a Raspberry Pi Polarization Camera and worked on RGB camera response function estimation from polarization cues

## System Architect, AMERGINT Technologies

May 2020–Mar 2021

- Advanced Systems *Colorado Springs, USA*
- Reorganized and modified the build system for faster performance – improved build speed by up to 60%

## App Development Intern, Keysight Technologies

May 2018–Aug 2018

- IT *Colorado Springs, USA*
- Built a cross-platform emergency awareness app that went into production using Xamarin (C# and XAML)

## Website Administrator, TEDxColoradoSprings / TEDxUCCS

Mar 2015–Mar 2016

- Web Development *Colorado Springs, USA*
- Built and maintained the TEDxColoradoSprings and TEDxUCCS websites

## Publications

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- [1] **SaiKiran Tedla**, Francesco Banterle, Trevor Canham, Karanpreet Raja, David B Lindell, Kiriakos N Kutulakos, Jiacheng Li, Feiran Li, and Daisuke Iso. “Generating HDR Video from SDR Video”. In: *ACM SIGGRAPH Asia Conference (SIGGRAPH Asia)*. 2026. [Project Page](#).
- [2] **SaiKiran Tedla\***, Joshua E Little\*, Hakki Can Karaimer, and Michael S Brown. “Off the Planckian Locus: Using 2D Chromaticity to Improve In-Camera Color”. In: *European Conference on Computer Vision (ECCV)* (2026). \*Equal contribution, **Spotlight Oral**, [Project Page](#).
- [3] **SaiKiran Tedla**, Abhijith Punnappurath, Luxi Zhao, and Michael S Brown. “Examining Joint Demosaicing and Denoising for Single-, Quad-, and Nona-Bayer Patterns”. In: *IEEE International Conference on Computational Photography (ICCP)*. 2025. [Paper Link](#).
- [4] **SaiKiran Tedla**, Zhoutong Zhang, Xuaner Zhang, and Shumian Xin. “Learning to Refocus with Video Diffusion Models”. In: *ACM SIGGRAPH Asia Conference (SIGGRAPH Asia)*. 2025. [Project Page](#).
- [5] **SaiKiran Tedla**, Kelly Zhu, Trevor Canham, Felix Taubner, Michael S Brown, Kiriakos Kutulakos, and David Lindell. “Generating the Past, Present, and Future from a Motion-Blurred Image”. In: *ACM Transactions on Graphics, SIGGRAPH Asia (ToG)*. 2025. [Project Page](#).
- [6] **SaiKiran Tedla\***, Junyong Lee\*, Beixuan Yang, Mahmoud Afifi, and Michael S Brown. “Multi-spectral Demosaicing via Dual Cameras”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025. \*Equal contribution, **Highlight award**, [Project Page](#).
- [7] **SaiKiran Tedla**, Scott MacKenzie, and Michael Brown. “LookToFocus: Image Focus via Eye Tracking”. In: *ACM Symposium on Eye Tracking Research and Applications (ETRA CO-GAIN)*. 2024. [Project Page](#).
- [8] **SaiKiran Tedla\***, Beixuan Yang\*, and Michael S Brown. “Examining autoexposure for challenging scenes”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2023. \*Equal contribution, [Project Page](#).
- [9] **SaiKiran Tedla**, Yunyuan Wang, Maitri Patel, and Michael S Brown. “Analyzing color imaging failure on consumer-grade cameras”. In: *Journal of the Optical Society of America A (JOSA A)* 39.6 (2022). [Paper Link](#).
- [10] Andrew Y Guo, Anagh Malik, **SaiKiran Tedla**, Yutong Dai, Yiqian Qin, Zach Salehe, Benjamin Attal, Sotiris Nousias, Kyros Kutulakos, and David B Lindell. “Dark3R: Learning Structure from Motion in the Dark”. In: *IEEE/CVF Computer Vision and Pattern Recognition (CVPR)*. 2026. **Highlight award**, [Project Page](#).

- [11] Hodaka Kawachi, Tomoya Nakamura, Yasushi Yagi, **SaiKiran Tedla**, Trevor Canham, and Michael Brown. “Embedding information into images taken with consumer camera using coded ambient lighting”. In: *IEEE/CVF Computer Vision and Pattern Recognition Findings (CVPRF)*. 2026. [Paper Link](#).
- [12] Trevor D Canham, **SaiKiran Tedla**, Michael J Murdoch, and Michael S Brown. “Gain-MLP: Improving HDR Gain Map Encoding via a Lightweight MLP”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025. [Project Page](#).
- [13] Abhijith Punnappurath, Luxi Zhao, Hoang Le, Abdelrahman Abdelhamed, **SaiKiran Tedla**, and Michael S Brown. “Improved mapping between illuminations and sensors for RAW images”. In: *Journal of the Optical Society of America A (JOSA A)* 42.8 (2025). [Paper Link](#).
- [14] Trevor Canham, **SaiKiran Tedla**, Michael J Murdoch, and Michael S Brown. “Gamma Maps: Non-linear Gain Maps for HDR Reconstruction”. In: *London Imaging Meeting (LIM)*. Vol. 5. Society for Imaging Science and Technology. 2024. [Paper Link](#).
- [15] Donghwan Seo, Abhijith Punnappurath, Luxi Zhao, Abdelrahman Abdelhamed, **SaiKiran Tedla**, Sanguk Park, Jihwan Choe, and Michael S Brown. “Graphics2RAW: mapping computer graphics images to sensor raw images”. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2023. [Paper Link](#).

## Preprints and Patents

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- [1] **SaiKiran Tedla**, Zhoutong Zhang, Xuaner Zhang, and Shumian Xin. “Learning to Refocus with Video Models”. Provisional patent application. 2026.
- [2] **SaiKiran Tedla**, Kelly Zhu, Trevor Canham, Felix Taubner, Michael S Brown, Kiriakos Kutulakos, and David Lindell. “Generating the Past, Present, and Future from a Motion-Blurred Image”. In: (2026). Provisional patent application.
- [3] **SaiKiran Tedla**, Abhijith Punnappurath, Luxi Zhao, and Michael Scott Brown. “Examining Joint Demosaicing and Denoising for Single-Bayer, Quad-Bayer, and Nona-Bayer Patterns”. US18823268. US Patent Application, Patent pending. May 2025. [Patent Link](#).
- [4] **SaiKiran Tedla\***, Junyong Lee\*, Beixuan Yang, Mahmoud Afifi, and Michael Brown. “Multi-spectral Demosaicing via Dual Cameras”. Provisional patent application, \*Equal contribution. 2025.

## Honors and Awards

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**Doctoral Consortium** 2026

*European Conference on Computer Vision (ECCV)*

- Accepted to the [ECCV 2026 Doctoral Consortium](#).

**Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship** 2026-2028

*Schmidt Sciences and University of Toronto*

- Awarded the [Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship](#) (\$85,000/year) in Dec 2025 to explore generative re-photography at the University of Toronto.

**Arts and Sciences Postdoctoral Fellowship (Declined)** 2026

*University of Toronto*

- Awarded the University of Toronto in Science postdoctoral fellowship (\$65,000/year) to explore generative re-photography at the University of Toronto.

**Outstanding Reviewer Award** 2025, 2026

*IEEE / CVPR*

- Recognized among the top 5.6% of reviewers for CVPR 2025 and top 5% for CVPR 2026.

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|----------------------------------------------------------------------------------------------------------------------|-----------|
| <b>PhD Graduate Fellowship</b><br><i>York University</i>                                                             | 2022-2026 |
| • Fellowship from York University (\$32,500/year).                                                                   |           |
| <b>People's Choice Award</b><br><i>Lassonde School of Engineering Photography Competition</i>                        | 2024      |
| • Received the People's Choice Award for best photograph in the annual Lassonde Engineering Photography Competition. |           |
| <b>Masters Graduate Fellowship</b><br><i>York University</i>                                                         | 2020-2022 |
| • Fellowship from York University (\$17,000/year).                                                                   |           |
| <b>Outstanding Undergraduate Student</b><br><i>University of Colorado Colorado Springs, Computer Engineering</i>     | 2019      |
| • Awarded to the top undergraduate student in Computer Engineering.                                                  |           |
| <b>Leon Young Scholarship</b><br><i>Educating Children of Color, Colorado Springs</i>                                | 2017      |
| • Awarded to the student who best honors the legacy of Leon Young, the first Black mayor of Colorado Springs.        |           |
| <b>Fairness for All Scholarship</b><br><i>Educating Children of Color, Colorado Springs</i>                          | 2017      |
| • Awarded to the student who best demonstrates fairness and justice within the community.                            |           |

## Talks

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|---------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| [1] Generative Re-Photography with Video Models<br><i>University of Washington, Hosted by Steve Seitz</i>           | May 2026<br><a href="#">Presentation</a> |
| [2] Generative Re-Photography with Video Models<br><i>CMU VASC Seminar, Hosted by Ioannis Gkioulekas</i>            | Apr 2026<br><a href="#">Presentation</a> |
| [3] Generative Re-Photography with Video Models<br><i>Princeton PIXL Seminar, Hosted by Felix Heide</i>             | Apr 2026<br><a href="#">Presentation</a> |
| [4] Generative Re-Photography with Video Models<br><i>Cornell Tech, Hosted by Noah Snaveley</i>                     | Apr 2026<br><a href="#">Presentation</a> |
| [5] Generative Re-Photography with Video Models<br><i>MIT CSAIL Vision Seminar, Hosted by Vincent Sitzmann</i>      | Apr 2026<br><a href="#">Presentation</a> |
| [6] Generative Re-Photography with Video Models<br><i>Harvard - Zickler Lab, Hosted by Todd Zickler</i>             | Apr 2026<br><a href="#">Presentation</a> |
| [7] Generative Re-Photography with Video Models<br><i>MIT Media Lab - Camera Culture, Hosted by Ramesh Raskar</i>   | Apr 2026<br><a href="#">Presentation</a> |
| [8] Generative Re-Photography with Video Models<br><i>Stanford University, Hosted by Gordon Wetzstein</i>           | Apr 2026<br><a href="#">Presentation</a> |
| [9] Generative Re-Photography with Video Models<br><i>University of California Berkeley, Hosted by Laura Waller</i> | Apr 2026<br><a href="#">Presentation</a> |
| [10] Generative Re-Photography<br><i>Sony AI, Tokyo, Japan</i>                                                      | Feb 2026<br><a href="#">Presentation</a> |
| [11] Generative Re-Photography<br><i>University of Osaka, Tokyo, Japan</i>                                          | Oct 2025<br><a href="#">Presentation</a> |
| [12] Generative Re-Photography<br><i>Spatial AI Network, Computer Vision League (CVL) – Osaka, Japan</i>            | Oct 2025<br><a href="#">Presentation</a> |
| [13] Examining Autoexposure for Challenging Scenes                                                                  | Mar 2024                                 |

## Reviewing

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- 1: Conference on Computer Vision and Pattern Recognition (CVPR), 2024–2026, **Outstanding Review Award 2025, 2026**
- 2: Conference on Neural Information Processing Systems (NeurIPS), 2026
- 3: SIGGRAPH Asia, 2026
- 4: European Conference on Computer Vision (ECCV), 2024, 2026
- 5: British Machine Vision Conference (BMVC), 2026
- 6: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025
- 7: SIGGRAPH, 2025
- 8: Winter Conference on Applications of Computer Vision (WACV), 2023

## In the Media

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|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| [1] Sai T. is helping advance the future of AI-powered photography and computer vision.<br><i>Lassonde School of Engineering, York University</i> | Jul 2026<br><a href="#">Link</a> |
| [2] Demand for computer scientists surging — not shrinking — as AI marches on<br><i>Toronto Star and York University</i>                          | Mar 2026<br><a href="#">Link</a> |
| [3] Colorado Springs senior earns bachelor's degree in high school<br><i>KRDO</i>                                                                 | May 2019<br><a href="#">Link</a> |

## Advising and Mentorship

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### Graduate Students

- [Karanpreet Raja](#) (Aug 2025 – Present)
- [Beixuan Yang](#) (Mar 2022 – Mar 2025)

### Undergraduate Students

- [Joshua Little](#) (Apr 2025 – Mar 2026)
- [Karanpreet Raja](#) (Jan 2025 – Aug 2025)
- [Rajat Ajaykumar](#) (Aug 2022 – Mar 2023)

## Teaching

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|------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>EECS 1516: Object-Oriented Design and Principles</b><br><i>Teaching Assistant, York University</i>      | Winter 2024, Winter 2025, Winter 2026 |
| <b>EECS 1022: Programming for Mobile Computing</b><br><i>Teaching Assistant, York University</i>           | Fall 2024                             |
| <b>EECS 1570: Computing for Psychology</b><br><i>Teaching Assistant, York University</i>                   | Winter 2023                           |
| <b>MATH 1090: Introduction to Logic for Computer Science</b><br><i>Teaching Assistant, York University</i> | Fall 2022                             |
| <b>EECS 1012/1015: Introduction to Computer Science</b><br><i>Teaching Assistant, York University</i>      | Fall 2021, Winter 2022, Fall 2023     |

## **Skills**

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**Programming:** Java, C++, C, Python, HTML, CSS, JavaScript, MATLAB

**Projects:** FastSwap (social media app for contact sharing via QR codes); Political Ideology Detector (NLP-based web app); Robotic Car (GPS navigation, obstacle avoidance, sensor integration); Quadcopters (reverse engineering and reconstruction)

**Certifications:** Oracle Java SE 8 Professional (OCP); C++ Associate (C++ Institute); CompTIA Linux+; LPIC-1: Linux Administrator

## **References**

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### **Dr. Michael S. Brown**

Professor, Dept. of Electrical Engineering and Computer Science

York University; Vice President, Samsung Research America; Director, Samsung AI Center Toronto  
Toronto, Canada

[Webpage](#)

Email: [michael.brown@yorku.ca](mailto:michael.brown@yorku.ca)

### **Dr. David B. Lindell**

Assistant Professor, Dept. of Computer Science

University of Toronto; Faculty Affiliate, Vector Institute  
Toronto, Canada

[Webpage](#)

Email: [lindell@cs.toronto.edu](mailto:lindell@cs.toronto.edu)

### **Dr. Kiriakos N. Kutulakos**

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